

Greenwich - Kids

UK Family Tour

Where the Whole World Starts Counting

Before you climb the hill, here is something amazing to know. Right here, in this one spot in London, two enormous things happened. A small, rainy island became the most powerful country in the whole world. And the entire planet decided to start counting time and place from this exact hill. You are about to stand on the spot where the world's map begins.

And here is the secret. Those two things are really one story. Britain is an island, with water on every side. It did not have gold mountains or deserts full of treasure. What it had was the sea. So the British built ships, thousands of them, and sent them everywhere, carrying tea, sugar, cotton and spices back and forth across the whole world. Their navy grew so strong that people said Britain ruled the waves, meaning it bossed the oceans. At its very biggest, the British Empire ruled about one quarter of all the land on Earth.

That sounds impressive, and it was. But it is also a hard story, because a lot of that power came from taking over other people's countries, and even from buying and selling enslaved people, which was terribly cruel. So today we will tell the whole truth, the amazing parts and the sad parts, both together.

To sail all the way around the world, though, British sailors first had to solve a puzzle that had stumped everybody for hundreds of years. When you are floating in the middle of the ocean, with water in every direction, how do you know where you are? The answer was figured out right here. So let's climb up and go find it.

How a Rainy Island Bossed the Oceans

Look at a map of Britain and you will spot something the British spotted long ago. It is an island, and no matter where you stand on it, the sea is never very far away. If you are a country that wants to become powerful, that is either a trap or a superpower. The British decided it was a superpower.

A long time ago, in fifteen eighty eight, a gigantic fleet of Spanish warships, called the Armada, sailed to invade England. But the English ships were smaller and faster, a huge storm blew up, and the invasion failed completely. After that, people started calling the navy Britain's wooden walls. Think about how clever that is. If your walls are made of ships, nobody can easily attack your island, but your walls can also sail away and turn up anywhere in the world. That is a very sneaky kind of power.

So British ships sailed everywhere. They took the things Britain made, and brought back tea from China, spices and cotton from India, and sugar from islands in the Caribbean. Everyone wanted these things, and trading them made Britain incredibly rich.

But here comes the hard part, and it matters. To grow all that sugar, British ships also carried enslaved people, kidnapped from Africa and forced to work with no freedom and no pay, in horrible conditions. It was one of the cruelest things in all of history. Britain finally banned it, first the slave trade in eighteen oh seven, and then slavery itself in eighteen thirty three, but only after a very long time and a lot of suffering.

Here is a number that is almost impossible to picture. At its biggest, the British Empire was so huge that the sun was always shining on some part of it, somewhere in the world, at every single moment of the day and night. That is why people used to say the sun never set on the British Empire. Just imagine that. No matter what time it was, it was daytime in some piece of it.

So as you walk around Greenwich today, you are really looking at two things at once. Amazing bravery, cleverness and adventure. And a powerful country that was sometimes very unfair to other people. Real history is like that. It is brilliant and messy at the very same time, and you are old enough to hold both.

The Puzzle That Sank a Thousand Ships

Imagine you are a sailor, hundreds of years ago, standing on a wooden ship in the middle of the ocean. Turn around. Water. Turn again. Water. There is no land, no signpost, no way to call home. Now answer one question. Where are you? Get it wrong, and you might sail for weeks in the wrong direction, run out of food and fresh water, or crash into rocks in the dark. This really happened, again and again, and thousands of sailors died because of it.

Half the puzzle was easy, even for the ancient Greeks. By looking at how high the sun climbed, or by finding the North Star at night, sailors could work out how far north or south they were. But how far east or west? That part seemed impossible. The sky gave no clue at all.

Then somebody had a brilliant idea. East and west are secretly about time. The Earth spins all the way around once every day, so the sun comes up earlier in some places and later in others. Here is the trick. If you carried a clock that always showed the time back home, and then you checked what time it was on your ship by watching the sun, the difference between the two clocks would tell you exactly how far around the world you had sailed. Clever, right?

There was just one problem, and it was a monster. You needed a clock that kept perfect time on a rocking, splashing, freezing ship. And in those days, even the best clocks, sitting still and safe in a warm room, could not do that. Everybody said it was hopeless.

But somebody did solve it, right here on this hill, and it took him most of his whole life. You will meet his amazing clocks at the top. For now, just think about how wild this is. To figure out where you are, people first had to figure out what time it is. Place and time turned out to be the very same puzzle. And because it was cracked here, this is where the whole world's map and the whole world's clock both begin.

Your Greenwich Treasure Hunt

Here is the plan for your day, and some things to hunt for as you go.

Start at the bottom, by the river, with the ship. She is the last of the great racing tea ships, and here is the wild part. They have lifted her right up into the air so you can walk underneath her and look up at her giant copper bottom. Almost nobody in all of history has stood under a sailing ship like that. You are about to.

Then you climb the hill. The big, beautiful, cream coloured buildings you walk through used to be a royal palace, and later a home for old and hurt sailors, built to say thank you to the men who sailed the ships. Keep climbing, and at the very top you reach the observatory, where clever people studied the stars, not just for fun, but to help sailors find their way.

Now, three things to find at the top, like a treasure hunt. Number one, a shiny metal line set into the ground. That is the start of the world's map, zero. Put one foot on each side and you will be standing in two halves of the planet at the same time. Number two, look up for a big, bright red ball on a pole. Every single day at exactly one o'clock it drops, and it has been doing that for almost two hundred years, so

ships could set their clocks by it. Try to catch the moment it falls. Number three, find the little machines in the glass cases. Those are the clocks that solved the giant puzzle and saved sailors' lives.

And here is a bonus. When you reach the top, turn around and look back down the hill. You will see beautiful white buildings, then the wide river, and then, across the water, a cluster of tall glass skyscrapers. Those towers stand on the exact spot where the old ships used to unload all that tea and sugar and cotton, hundreds of years ago. So in one single view you can see the whole story. Old sailing ships, then the river, then the shiny modern city that grew out of it all. See if you can spot all three.

One island, one hill, and the beginning of the whole world's map and clock. Come on, let's climb.

Cutty Sark: The Fastest Ship of Her Day

You're standing under a real racing ship — one of the fastest sailing ships ever built. The Cutty Sark was launched in 1869 to do one job: race tea from China to London. Back then, the first ship home with the new tea crop got the best prices, so tea clippers raced halfway around the world, and newspapers reported it like a sport. Cutty Sark could sail over three hundred and sixty miles in a single day, faster than many early steamships. Her strange name comes from a Scottish poem about a witch named Nannie who chased a farmer while wearing a short nightshirt — a “cutty sark” in Scots. Look at the figurehead at the front of the ship: that's Nannie, reaching out. In the poem, she grabbed the tail of the farmer's horse as he escaped, so sailors put a real horse's tail in her hand for luck. Later, the ship raced wool from Australia and beat every rival. In 2007, while being repaired, she caught fire — but most of her original wood and iron had been removed for cleaning, so she survived. Walk right underneath her shiny copper bottom. Very few people in history have stood under a ship like this.

Related listening: Turner painted sail giving way to steam; Cutty Sark lived that same ending.
Coming up — “The Fighting Temeraire: A Hero's Last Trip” (National Gallery, Day 5).

Life Aboard a Tea Clipper

Now imagine you actually lived on this ship. You'd be a boy not much older than you are, an apprentice, learning the ropes, and there were a lot of ropes. Cutty Sark had eleven miles of rigging, and the sailors had to know every single line in the dark, in a storm, half asleep. When the captain shouted for more sail, up you went, climbing the mast while the ship rolled and the wind screamed, sometimes a hundred and fifty feet above a freezing sea. There was one rule the sailors lived by. One hand for yourself, one hand for the ship. It meant always keep one hand gripping something solid, because the ocean does not give you a second chance.

The food would horrify you. Fresh meat lasted only a few days, so most of the voyage you'd eat hard biscuit and salt beef from a barrel. The biscuit was so tough sailors called it hardtack, and here's the yuck part. It was often full of little beetles called weevils. The trick was to tap your biscuit on the table so the wriggling things fell out first, or just eat in the dark so you couldn't see them.

And there was a hidden danger worse than storms. On long voyages, sailors' gums would bleed, their teeth would fall out, and old wounds would open up again. This was a disease called scurvy, and it killed more sailors than shipwrecks did. Nobody understood it for centuries. Then people worked out that fresh fruit, especially lemons and limes, kept it away. British ships started carrying lime juice for the crew to drink, which is why, all around the world, British sailors got the nickname limeys. It's still used today.

So next time you look up at those towering masts, picture a boy your age up there in the rigging, one hand for the ship, a stomach full of weevil biscuit, waiting for the sight of land.

The Great Rudder Rescue

Here's a true adventure that sounds made up. In 1872, Cutty Sark had a deadly rival, another beautiful clipper called Thermopylae, and the two ships left China on the very same day to race each other home to London. For weeks they were neck and neck across the whole Indian Ocean. Then Cutty Sark pulled ahead, four hundred miles in front. Victory looked certain.

And then disaster. A huge gale smashed into the ship, and with a terrible crack the rudder broke clean off and vanished into the sea. The rudder is the flat blade at the back that steers the whole ship. Without it, Cutty Sark could only spin helplessly, like a car with no steering wheel, thousands of miles from any port.

The safe choice was to limp to the nearest harbor and give up the race. The captain, George Moodie, refused.

Instead, the ship's carpenter, a man named Henderson, decided to build a brand new rudder out of spare timber and iron, at sea, in a storm. But you can't shape iron without fire, so they needed a blacksmith. Here's the amazing part. When the ship had left China, they'd discovered two stowaways hiding on board, and one of them just happened to be a blacksmith. So this stowaway set up a little forge right there on the heaving deck, flames and all, while giant waves crashed over them. At one point the burning coals spilled and badly scorched the captain's own son. It took six days of brutal, dangerous work, but they built that rudder and bolted it on and sailed on.

Cutty Sark lost the race by only a week, after crossing half the planet with a homemade rudder. Everyone agreed it was one of the bravest feats in the age of sail. Now, before you leave, look in the gallery beneath the ship for rows of carved wooden figureheads, more than a hundred of them, rescued from lost ships by a collector with an eye patch who called himself Long John Silver.

The Prime Meridian: Stand in Two Halves of the World

At the top of the hill, there's a metal line in the courtyard, and it might be the most famous line on Earth. It's the Prime Meridian — zero degrees longitude — the line that divides the Eastern half of the world from the Western half. Put one foot on each side. Your left foot is in the same half of the world as America; your right foot is with China and Australia. You are standing in both halves of the planet at once. Why here, in Greenwich? Because sailors needed a starting line to measure where they were, and in 1884, delegates from twenty-five countries met in Washington and voted to make Greenwich the world's zero. Most of the world's ships were already using Greenwich maps, so it won easily. France voted no and grumpily used its own Paris line for years afterward. This is also why world time is called Greenwich Mean Time — every time zone on Earth is counted from this exact spot. When someone in Texas says it's 7 AM Central Time, they really mean “six hours behind this courtyard.” Watch for the red ball on the roof: it drops at exactly 1 PM every day, and it has since 1833, so ship captains on the river could set their clocks.

The Line That Circles the World

That metal line under your feet doesn't stop at the edge of the courtyard. Imagine it stretching out in both directions and keep going in your mind. It runs north across England, over the sea, all the way up to the North Pole. And it runs south too, down through France and Spain, across all of Africa, over the ocean, right to the South Pole. It's an invisible circle drawn around the whole planet, and you are standing on it.

At night, the Observatory does something magical to show you. It fires a bright green laser beam from the hill, pointing north, following the exact line of the meridian across the London sky. On a clear evening you can see it for miles, a thin green thread hanging over the city, marking zero.

Now here's a puzzle to twist your brain. If this line is zero, what's on the exact opposite side of the Earth? Dig straight down through the middle of the planet and come out the other side, and you'd be in the Pacific Ocean, far from any land. And running through that empty ocean is another special line, called the International Date Line. It's where today turns into tomorrow. Step across it going one way and you jump forward a whole day. Step back and you're in yesterday again. Sailors have crossed it and celebrated the same birthday twice, or missed a day completely.

So this quiet line in a Greenwich courtyard has a twin on the far side of the world where the date flips over. Everything starts here. The map makers had to pick somewhere to call zero, the same way a race needs a starting line, and they picked this hill. Put your feet on the line one more time. To the east, the whole world is a little bit ahead in time. To the west, it's a little behind. You are standing exactly in the middle of the world's clock.

Why Your Phone Disagrees

Try something clever right here. If a grown-up has a phone with a maps app, ask them to find the spot where it says zero degrees longitude. Then watch their face, because the phone will not point at this famous metal line. It will insist that zero is over there, about a hundred meters to the east, near some bushes in the park. So who's wrong, the phone or the line that the whole world agreed on in 1884?

Here's the surprise. Nobody is wrong. They're just measuring in two different ways. The old line was set by astronomers looking straight up at the stars, using gravity and a hanging weight to find true vertical. But the Earth isn't a perfect ball. It bulges and dips, and the ground under Greenwich actually tugs a

weight very slightly sideways. Your phone uses satellites spinning in space instead, and they measure from a smooth, imaginary, perfectly even Earth. Those two methods disagree by about a hundred meters. So there are, in a funny way, two zeros, and you can walk from one to the other in a minute.

Now think about what this line really does for everybody. Every clock on Earth is counted from right here. When it's noon on this hill, ask yourself what time it is at Grandma's house, or in Australia, or in China. Each place is a set number of hours ahead or behind Greenwich. That's what a time zone is, a slice of the world that shares the same time, all measured from this one spot.

It's also the reason for that groggy, upside-down feeling called jet lag. When you fly a long way east or west, you cross those slices too fast. Your body still thinks it's back home, ready for breakfast, while outside it's the middle of the night. Your insides are still running on Greenwich, so to speak, and they take a few days to catch up. All of it, every time zone and every bit of jet lag, begins on this line beneath your shoes.

John Harrison: The Carpenter Who Saved Sailors

Inside the Observatory are four of the most important machines ever built — and they were made by a carpenter with no training in clocks. Here's the problem they solved. For centuries, sailors could tell how far north or south they were by the sun and stars. But east or west? No idea. Ships got lost, ran out of food, and smashed into rocks. In 1707, four British warships hit rocks in fog and about two thousand sailors drowned — because the fleet didn't know where it was. So in 1714, Parliament offered a gigantic prize — twenty thousand pounds, like millions today — to anyone who could solve “the longitude problem.” The trick was time: if you know the exact time back home and compare it to noon where you are, the difference tells you how far east or west you've traveled. But no clock could keep time on a rolling, damp, freezing ship. Everyone said it was impossible. John Harrison, a self-taught village carpenter, spent over forty years building clock after clock — H1, H2, H3, each one weirder and more wonderful — until H4, which looks like a big pocket watch, kept nearly perfect time across the Atlantic. The fancy scientists didn't want to pay a carpenter, and it took the King himself to step in. Find H4 in its case. That little watch changed the world.

The Boy Who Taught Himself

Let's go back to when John Harrison was a boy, because his whole story is a wonderful underdog tale. He grew up in a country village and his father was a carpenter, so John learned to work with wood. There was no school for clockmaking, no teacher, no fancy college. Everything he knew, he taught himself.

Here's a lovely story people tell about him. When John was six years old and sick in bed, someone gave him a watch to keep him company, and he lay there for hours just listening to it tick and watching the tiny wheels spin. He was hooked for life. By the time he was twenty, this self-taught carpenter had built a complete working clock, and here's the astonishing part. He built almost the whole thing out of wood. The gears, the wheels, all carved from timber.

Now, wooden clocks have a problem. Metal gears need oil to stop them grinding away, but oil gets thick and gloopy and stops working. So young Harrison had a brilliant idea. He used a special hard, oily wood called lignum vitae that is naturally slippery. His wooden clocks never needed a single drop of oil. Some of them are still ticking today, three hundred years later. Imagine building something so good it outlives you by centuries.

He also invented a clever part shaped a bit like a grasshopper's kicking legs, which gave each swing a gentle push and made almost no rubbing at all. Clever, quiet, and it barely wore out.

So picture him. No money, no teachers, no famous name, just a carpenter in a village who understood time better than anyone alive. And he was about to take on the smartest scientists in the whole country, who were absolutely certain that a mere carpenter could never solve the greatest puzzle of the age. They were about to be proved very, very wrong.

Clock Versus Moon

Harrison had a rival, and it wasn't another clockmaker. It was the Moon. A group of clever astronomers were certain that the way to find your position at sea was to measure the Moon against the stars. As the Moon drifts across the sky, it slowly slides past the stars behind it, a bit like the hand of a giant clock in space. If you measured that very carefully and did pages and pages of tricky sums, you could work out the time back home, and from that, where you were. It worked, but it was slow and hard, and on a cloudy night you got nothing at all.

The man in charge of testing both ideas was the Astronomer Royal, Nevil Maskelyne, and here was the problem. Maskelyne loved the Moon method. He had written the very book sailors used to do the sums. So when it came time to judge whether Harrison's clock was better than the Moon, the judge was cheering for the other team. Harrison felt this was terribly unfair, and honestly, he had a point.

But the clock had one huge advantage. Anybody could use it. You didn't need to be a genius astronomer or do hours of maths in a rolling ship. You just looked at the time. And when they finally tested Harrison's watch on real voyages across the ocean, it worked beautifully.

The proof that won everyone over came from a copy. A craftsman built an exact copy of Harrison's watch, and it was carried by the famous explorer Captain James Cook on his great voyage around the world. Cook used it to map coastlines with amazing accuracy and came to love it, calling it his trusty, never-failing friend. After that, no one could argue. Ships everywhere began carrying a clock set to Greenwich time, and sailors were never hopelessly lost again. The self-taught carpenter had beaten the Moon.

Nelson's Coat and a Ball That Tells Time

Two more treasures before you leave Greenwich. First, look up at the Observatory roof for the bright red ball on a pole. Every day at 12:55 PM it rises halfway, at 12:58 it rises to the top, and at exactly 1 PM it drops. It's been doing this since 1833. Why? So captains of ships on the Thames could look up the hill, see the ball drop, and set their ship's clocks before sailing — remember, exact time was how they found their way at sea. It was basically the world's first public time signal, long before radio. Down the hill in the National Maritime Museum — which is free — hunt for one dramatic object: the actual uniform coat Admiral Lord Nelson was wearing when he was shot at the Battle of Trafalgar in 1805. You can see the bullet hole in the left shoulder. Nelson was Britain's greatest naval hero, and he won the battle but died that day; you'll see his column and statue tomorrow in Trafalgar Square, guarded by four giant lions. And one more thing to notice: the museum has a huge ship in a bottle — a real-sized model of Nelson's ship *Victory*, with patterned sails, inside a giant glass bottle. How did they get it in there? Come look and make your guess.

Related listening: Christopher Wren designed both the cathedral and Greenwich's riverside masterpiece. Listen back — “St Paul's: The Dome That Refused to Burn” (St Dunstan & St Paul's, Day 3). Related listening: Nelson's coat is at Greenwich; his column and lions preside over Trafalgar Square. Coming up — “Pelicans, Lions and a One-Armed Hero” (Westminster Walk, Day 6).

The Ceiling That Took a Lifetime

Down the hill there's a room so beautiful that people call it the Sistine Chapel of Britain. It's called the Painted Hall, and when you walk in, tip your head all the way back and look up, because every inch of that gigantic ceiling and the walls is covered in painting. Kings, queens, angels, ships, all swirling above you in gold and blue.

Now here's the wow. Almost all of it was painted by one man, an artist called James Thornhill, and it took him nineteen years. Nineteen years, mostly lying on his back or standing on high wooden scaffolding, dabbing away at a ceiling far above the floor. Imagine starting a drawing and only finishing it when you're nearly an old man. He got paid by the amount he painted, so much for each patch of ceiling, so much for each patch of wall, and there's a cheeky legend that near the edge he painted his own hand reaching out, as if he's still asking to be paid.

This grand hall was built as a dining room for old and injured sailors, retired from the navy, who lived here in what was basically a huge home for sea veterans. But the room turned out so posh and splendid that the sailors weren't allowed to eat their ordinary meals in it. Too grand for dinner.

It had one very solemn moment though. When Admiral Nelson, Britain's great hero, died at sea, his body was brought back and laid here in this hall so that thousands and thousands of people could file past and say goodbye. That was in January, 1806. The crowds were so huge it was almost overwhelming.

And here's a fun modern twist. Because it's so spectacular, film crews love it. This hall and the buildings around it have appeared in *Pirates of the Caribbean*, in *Thor*, and in shows like *Bridgerton*. So the room where sailors once dined has become one of the most filmed spots in all of Britain. See if it looks familiar.

The Ghost on the Spiral Stairs

There's an elegant white building near the river called the Queen's House, and it hides one of London's most famous ghost stories. Inside is a beautiful spiral staircase that curls up and up with no central pole

holding it, seeming to float in mid air. It's called the Tulip Stairs, because of the flowers in the ironwork, and it was the first staircase of its kind ever built in Britain, over four hundred years ago.

Here's the spooky part. The story goes that in 1966, a visiting minister took an ordinary photograph looking up the stairwell. There was nobody there. But when the picture was developed, it seemed to show a shadowy figure, maybe even two, gripping the railing and climbing the steps, as if wrapped in a misty cloak. People have argued about that photograph ever since. Was it a trick of the light? A double exposure? A real ghost? Nobody has ever fully explained it, and to this day some visitors say the staircase feels strange and cold. When you climb it, keep one eye out, and see how you feel.

The Queen's House is full of treasures too. Look for a stunning painting of Queen Elizabeth the First, standing proud with her hand resting on a globe of the world, and ships battling a storm behind her. It celebrates the time England beat a giant invading fleet called the Spanish Armada. Her hand on the globe is basically saying, the world is mine.

And here's something to make this whole place feel real. Long before any of these buildings existed, there was a royal palace right here on this spot, and two of England's most famous rulers were actually born in Greenwich. Henry the Eighth, the king with six wives, was born here. And so was his daughter, that same Queen Elizabeth from the painting. So as you wander the lawns, remember, a king and a queen once took their first breath on this very ground, hundreds of years ago.

How Sailors Found Their Way: The Puzzle of Longitude

Think about the Cutty Sark you saw today, sailing all the way from China to London across an ocean with no roads, no signs, and no land in sight for weeks. How on earth did the captain know where he was? The answer is one of the greatest puzzles in history, and people died before they solved it.

To find your place on the sea, you need two numbers. The first is how far north or south you are, called latitude. That one is easy, and sailors have known how to find it for thousands of years. At noon you measure how high the sun is in the sky, or at night you look at the Pole Star, which sits almost exactly over the North Pole and hardly moves. From that you can work out your latitude. Easy.

The second number is how far east or west you are, called longitude, and this one was a nightmare. There is no special star for east and west, because the whole sky slides past as the Earth spins. For hundreds of years, nobody could figure it out, and ships got horribly lost. Once, in 1707, a whole British fleet thought it knew where it was, but it was wrong, and it smashed onto the rocks in the fog. Around two thousand sailors drowned in a single night.

Here is the clever secret that finally cracked it. Longitude is really about time. The Earth turns all the way round once a day, so as it spins, noon happens at different moments in different places. If you have a clock set to the time back home in London, and you look up and see that your own noon, when the sun is highest where you are, happens four hours later than your London clock says, then you know exactly how far around the world you have travelled. Time is the key to place.

But there was a huge problem: no clock could keep the right time on a ship. The best clocks used a swinging pendulum, and a swinging pendulum goes crazy on a rocking, rolling boat. Heat and cold and salty sea air ruined them too. A clock that was wrong by even a tiny bit each day would be wildly wrong after two months at sea, and that meant shipwrecks.

So the King offered a giant prize, a fortune, to anyone who could solve it. And the hero who did was not a famous scientist but a quiet carpenter named John Harrison, who taught himself to build clocks. He spent forty years, almost his whole life, building sea clocks that would not lose time no matter how the ship tossed. His fourth one, which looks like a giant pocket watch, was amazing: on a long voyage it lost only about five seconds. You can go up the hill to the Observatory and see his actual clocks still ticking today.

Poor Harrison almost didn't get his prize, because some powerful astronomers wanted the reward for their own method and kept making him prove it over and over. He was an old man before the King himself stepped in and said Harrison should be paid. The famous explorer Captain Cook took a copy of Harrison's clock all the way around the world and called it his trusty friend, because it always got him home.

By the time the Cutty Sark was racing tea across the sea, every ship carried one of these special clocks, kept safe below deck. Every single day the officers measured the sun, checked their clock, and marked a little dot on the map: this is where we are. That daily routine is how they crossed oceans and won races.

And that is why the whole world sets its maps and clocks from Greenwich. Somebody had to be the starting line for measuring east and west, and because British ships and star-charts were the best, in 1884 the countries of the world agreed: Greenwich would be zero. So when you stand on the brass line in the courtyard with one foot in the east and one in the west, you are standing on the spot where the whole planet begins counting. Not bad for a small hill in London.

The Story of Time: From Shadows to Atoms

Everything you saw at Greenwich today was really about one big question: what time is it, actually? That red ball that drops at one o'clock, the famous words Greenwich Mean Time, the room full of clocks, all of it is part of a story that took thousands of years to figure out. And it is a brilliant story, so let me tell it from the very beginning.

For almost all of history, time meant the sun. The very first clock was a shadow. Long ago in Egypt, people noticed that a stick in the ground casts a shadow that swings around during the day, short at noon when the sun is high, long in the morning and evening. That is a sundial. It works beautifully, until a cloud comes over or the sun goes down, and then you are stuck. So people invented clocks that did not need the sun, like water clocks that dripped drop by drop, and even clocks made of burning candles marked with lines. But all of them drifted and none could be trusted for long.

The big breakthrough came from monks. In their monasteries, they needed to know when to ring the bells for prayers, so about seven hundred years ago clever people built the first machines that kept time using heavy falling weights and a clicking wheel. These early clocks had no hands at all, they just rang a bell, and that is actually where our word clock comes from, an old word for bell. They were not very accurate, but they were magic, because for the first time a machine could keep time all through the night.

Then, about three hundred and fifty years ago, a scientist realised that a swinging weight, a pendulum, keeps amazingly steady time, tick, tock, tick, tock. Clocks with pendulums, like tall grandfather clocks, suddenly became a hundred times better. But a pendulum has to hang still, so it was useless on a rocking ship, which is exactly why Harrison had to invent his special spring-powered sea clocks that you heard about this morning.

Now here is the surprising part. Even with wonderful clocks, for a very long time there was no such thing as THE time. Every town set its own clocks by its own sun, so noon in one town was a few minutes different from noon in the next town along. Nobody minded, because people travelled slowly. Then along came the railways, and fast trains cannot run on a muddle of different times, or they might crash. So the trains made everyone use one single time, and they chose the time kept right here at Greenwich. That is why the red ball drops at one o'clock: so ships on the river could set their clocks to the exact same moment.

Later the whole world agreed to do the same thing. In 1884 the countries met and decided to measure all the world's time zones starting from Greenwich. That is why, wherever you are on Earth tonight, your time is counted as so many hours ahead of or behind Greenwich. This little hill became the place where the whole world starts counting time.

But the story keeps going. People discovered that even the Earth is not a perfect clock, because it wobbles and spins ever so slightly slower over the years. So scientists built the most incredible clock of all: the atomic clock, which counts the tiny buzzing of atoms, more than nine billion buzzes every second, and never goes wrong. Today this atomic time runs the whole modern world. Here is the coolest example: the satellites that help a phone find where you are work entirely by super-exact time. Each satellite has an atomic clock, and your phone works out your position by measuring, to billionths of a second, how long the signals take to arrive.

So when you watched that ball drop over Greenwich today, you were seeing the tip of an idea that runs all the way from a shadow on the ground to an atom humming in a lab. From the sundial, to the monks, to the pendulum, to Harrison, to the railways, to Greenwich giving the world its starting line, to the atomic

clock, humans slowly learned to measure time so perfectly that we barely need the sun at all. And it all comes back to this one small hill in London.

Related listening: From Greenwich's master clocks to Britain's most famous public clock.
Coming up — "Big Ben: The Most Famous Bong in the World" (Westminster Walk, Day 6).